

Cognitive Interactive Text Learner (CITL) Assistant - AI Study Companion

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Abstract - *The Cognitive Interactive Text Learner (CITL) is an android application with AI assistant designed for studying and interacting with a PDF document module to address the challenge in learning extensive academic materials such as PDF document module, PowerPoint slides and textbooks. The application secures any document through its offline LAN AI server (OLLAMA) that enables the application to process data locally without the use of the internet. The system integrates PDF text extraction, automated summarization, interactive flashcards and quizzes, and context aware AI chat enabling active interaction with the user reducing cognitive load. This platform utilizes the power of technology to support students to learn and overcome cognitive challenges, it supports the innovation in learning and technology.*

Keywords: *Cognitive, Learning, OLLAMA, Offline AI, Text Learner, PDF, Interactive, AI assistant, Android, Offline AI, Study companion, Local server.*

INTRODUCTION

The learning development process of students faces challenges not just because of the lessons but because of the resources available and the environment. Many students face severe “digital divide” because of unequal access to a stable internet connection and limited access to premium textbooks and online platforms. When learning materials are abundant, they are often delivered as unorganized digital “thick books” often characterized as “lengthy textbooks” so students spend more time navigating technical reading and difficulty extracting the core information^[1].

Mobile learning emphasizes that studying outside the traditional classroom often exposes students to an undisciplined learning environment distracting their learning, affecting their learning to a poor growth. Providing students with personal, adaptive tools allows them to block out environmental chaos by engaging directly in self-paced, active learning workflows ^[2]. Traditional learning from dense academic documents like textbooks and detailed PDF slides results in a significant cognitive challenge for students. Academic literature highlights that static digital documents require comprehensive reading, frequently leading to information overload, passive reading habits, and lower retention rates^[3].

These cognitive learning challenges are currently being faced by the students at Central Bicol State University of Agriculture - Sipocot Campus particularly the students in college of information technology where all learning resources are disseminated digitally. Digital document (PDF) files are used for learning modules, these files contain comprehensive topics about specific lessons in programming, database management, and other learning focus that needs a detailed discussion of every topic. Students having difficulty in learning due to information overload resulting in poor cognitive performance, for the students to improve their learning capabilities they need a platform that will extract the core information of the learning materials to the basics, fundamentals and information needed to learn specific lessons.

Educational psychology studies firmly establish that active recall and self-paced testing are significantly superior to passive rereading for long-term knowledge retention. Recent advancements in Natural Language Processing (NLP) and Large Language Models (LLMs) have enabled the automation of these strategies. Advanced features such as Summarization and AI chat changes the passive reading to a dynamic context-based questioning, these features allow the student to improve their comprehension and learning^[4].

The students are using a cloud-based learning platform to learn but the problem is their learning is not that secure due to the environment of AI, all the information you uploaded online will be learned by the cloud platform they are using. According to Ilarionov et. al. cloud-based learning offers limitations regarding data privacy, security, subscription costs, and internet dependency. To combat these challenges the students need an offline AI eco system to process data locally without uploading it to the cloud offering the students secure and reliable data privacy about their learning.

To handle these challenges the CITL: Cognitive Interactive Text Learner AI Assistant offers an android application with core features such as PDF extraction, data summarization, content-aware questioning, offline AI, interactive flashcards and quizzes, and performance monitoring. These features enable students to learn interactively with the system, lessening the burden in studying instead of reading entirely the comprehensive digital resources. This application aims to improve the cognitive learning of the students at Central Bicol State University of Agriculture - Sipocot Campus and to provide them a platform to use AI assistants with security and data privacy through a LAN based AI model.

OBJECTIVES

The primary objective of this study is to develop CITL: Cognitive Interactive Text Learner AI Assistant an android mobile application system for interactive learning through a LAN-based offline AI model using OLLAMA providing the users advanced platform for learning. This is designed to address the cognitive learning challenges faced by students at Central Bicol State University of Agriculture – Sipocot Campus, particularly those in the College of Information Technology. The application aims to:

1. To develop an Android application that would facilitate the learning experience by enabling the users to leverage AI locally through the use of Ollama.
2. Creation of a mobile-based interactive learning platform that allows for active learning, including recall exercises and context-based questions, without any dependencies on cloud APIs.
3. Automatic creation of customized learning aids in the form of summaries, multiple choice questions, and interactive quizzes and flashcards from uploaded PDFs.
4. To evaluate the effectiveness of the CITL Assistant through technical performance testing and user acceptance testing to measure its impact on study efficiency.

TARGET USERS

The primary target users of this application are the students at Central Bicol State University of Agriculture – Sipocot Campus, particularly within the College of Information Technology; these students rely heavily on digital resources such as PDF modules, lectures slides, and textbooks. While these learning resources are very comprehensive and detailed specifically in programming syntax and database queries they often present challenges such as information fatigue, passive reading habits, and difficulty in extracting core features.

These users face additional barriers to the digital divide, including limited access to stable internet connections and premium learning platforms. Providing features such as PDF text extraction, automated summarization, content-aware questioning, offline AI, interactive flashcards and quizzes, and performance monitoring, the application directly addresses the cognitive challenges students encounter in their academic journey.

SCOPE AND DELIMITATIONS

The CITL: Cognitive Interactive Text Learning AI Assistant mobile application is designed to assist students in their academic learning through the use of a LAN-based API server utilizing OLLAMA as AI model, the application covers the following modules:

PDF extraction - it extracts the text in the PDF document using Retrieval-Augmented Generation (RAG) to process and provide interactive learning.

Automated summarization - Summarize the data of the document and provide its core information needed by the student.

Content-aware questioning - generate answers that are directly based on the document and the questions asked by the user.

Offline AI: allows the application to process data locally using OLLAMA API.

Interactive quiz: generate questions based on the document uploaded and provide interactive UI for answering.

Generate flashcards: generate information based on the document and provide interactive flip cards with key terms and definitions.

Performance monitoring: track performance for each document uploaded and monitor learning.

While the system addresses key challenges in learning process it has the following limitations:

Cloud-based API: the application will not use any cloud-base API for data privacy and security.

Admin dashboard: the app runs independently for assisting student learning.

Scalability: the application is tailored for the needs of the students in CBSUA-Sipocot and may require future adjustments to accommodate multiple institutions.

File handling: the application handles PDF files only, it does not allow other file formats.

METHODOLOGY

The development of the CITL AI Assistant will utilize the Rapid Application Development (RAD) methodology. This approach shortens the development lifecycle by focusing on rapid development and iterative refinement instead of rigid documentation.

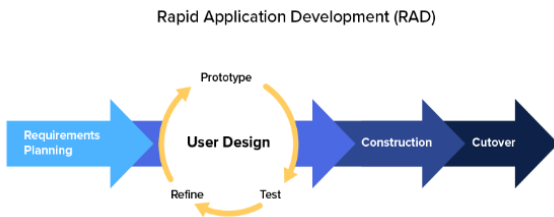


Figure 1 Rapid Application Development (RAD)

1. Requirements Planning Phase

In this initial stage, the researchers identified the core functional requirements: PDF text extraction, localized AI processing via Ollama, and the generation of active recall aids (quizzes and flashcards). Unlike traditional models, requirements remain flexible to account for the evolving capabilities of the local LLM integration.

2. User Design Phase (Iterative Prototyping)

During this phase, the researchers will develop functional prototypes of the Android interface. Users (students) will interact with early versions of the PDF viewer and AI chat.

Prototyping Focus: Testing the "Material 3" UI components and the "Stop" button logic to ensure the user experience is intuitive.

Feedback Loop: Feedback from these sessions will immediately inform the next design iteration, ensuring the tool actually reduces cognitive burden as intended.

3. Construction Phase

This phase involves the heavy lifting of software engineering, where prototypes are converted into the final product.

AI Integration: Implementing the Retrieval-Augmented Generation (RAG) pipeline, chunking PDF text into a Room Database, and establishing the Retrofit streaming connection to Ollama.

Rapid Coding: Using Kotlin Coroutines and View Binding to accelerate development while maintaining a clean, modern codebase.

4. Cutover Phase

The final phase includes technical performance testing and user acceptance testing (UAT). The application is finalized for its "first release." Documentation will be completed, and the installation/setup guide will be verified against real-world hardware configurations.

USER ACCEPTANCE TEST (UAT)

The User Acceptance Test (UAT) was conducted to evaluate the usability and overall user experience of the CITL Assistant mobile application. The evaluation was anchored on the ISO 9241-11 standard, which assesses usability across three primary dimensions: Effectiveness, Efficiency, and Satisfaction.

A structured questionnaire was distributed to a sample of 25 target users. The evaluation tool utilized a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree) to quantify the participants' feedback across 14 specific criteria. The collected data was tabulated and analyzed by calculating the mean scores to determine the application's impact on reducing cognitive load and improving study efficiency.

RESULT AND DISCUSSION

The application consists of several core modules that serve as the tool for learning and interacting. Its core features are PDF text extraction, automated summarization, interactive quiz and flashcards. The following figures shows how the functionalities work.

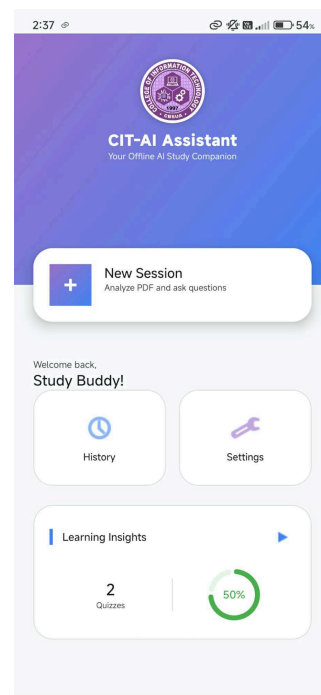


Figure 2 CITL Dashboard

This is the landing page of the system wherein the user can choose the features to be used. It allows the user to create a new session for interacting with the PDF file.

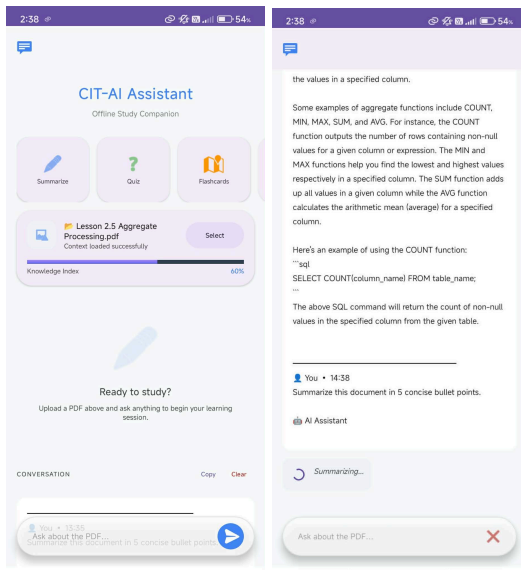


Figure 3 Automated summarization

This allows the users to attach a PDF file to be extracted and processed by the application and the system will summarize the information based on its content.

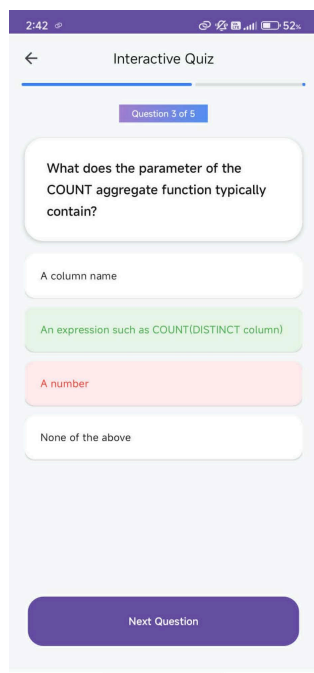


Figure 4 Interactive Quiz

This figure shows the feature of interactive quiz, it enables the user to create a quiz using the content of the PDF attached and answer the questions, it also records the score of the quiz to track the performance.

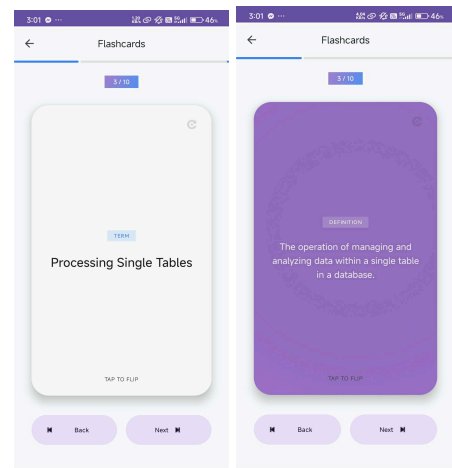


Figure 5 Interactive Flashcards

This figure shows the feature interactive flipcards: it generates a terms and definitions base in the attached PDF and processes it to create an activity user interface to interact with the user.

The results of the User Acceptance Test (UAT) are categorized according to the three ISO 9241-11 usability metrics. The numerical data reflects the participants' acceptance of the CITL Assistant as an offline AI study companion.

Category	Question	Mean Score (out of 5)	Overall Category Mean
Effectiveness	Q1. Features sufficient for PDF needs	4.88	4.79
	Q2. AI quiz generation relevant	4.76	
	Q3. Helpful messages for errors	4.80	
	Q4. Easy to recover from mistakes	4.72	
Efficiency	Q5. Icons/labels easy to understand	4.68	4.62
	Q6. Know how to use 'Summarize'	4.68	
	Q7. Little time to understand AI interaction	4.16	
	Q8. Easy navigation between PDF/chat	4.84	
	Q9. Control of pace and sequence	4.76	

Satisfaction	Q10. Explore features without fear	4.80	4.76
	Q11. App behaves consistently	4.68	
	Q12. Vocabulary easy to understand	4.84	
	Q13. Visually appealing interface	4.80	
	Q14. AI active recall is motivating	4.68	

1. **Effectiveness** (Overall Mean: 4.79) Effectiveness measures the accuracy and completeness with which users achieve specific goals using the application. Based on the evaluation, the application's features were deemed highly sufficient for PDF study needs, yielding an outstanding mean score of 4.88. Furthermore, users found the AI quiz generation to be relevant to the document content (Mean = 4.76). The application also demonstrated robust error handling, with users noting that it provided helpful messages (Mean = 4.80) and allowed for easy recovery from mistakes, such as selecting the wrong file (Mean = 4.72). The high effectiveness scores indicate that the offline OLLAMA-based RAG architecture successfully extracts and processes core information without relying on cloud-based APIs.

2. **Efficiency** (Overall Mean: 4.62) Efficiency assesses the resources and effort expended by users to achieve their goals. The interface design proved highly intuitive; participants agreed that navigation between the PDF viewer and the AI chat was seamless (Mean = 4.84) and felt in control of the pace and sequence of their interactions (Mean = 4.76). Users easily understood the icons and labels (Mean = 4.68) and immediately knew how to utilize the 'Summarize' feature (Mean = 4.68). Notably, the lowest score in the survey pertained to the learning curve of interacting with the AI (Mean = 4.16). While still a positive score, it suggests that first-time users may require a brief onboarding tutorial to fully grasp how to prompt the offline AI model. Overall, the results suggest that the application successfully mitigates "information fatigue" by providing a streamlined, self-paced learning workflow.

3. **Satisfaction** (Overall Mean: 4.76) Satisfaction evaluates the users' comfort and positive attitudes toward the use of the system. The CITL Assistant encouraged users to explore its features without the fear of making mistakes, obtaining a strong mean score of 4.80. Participants appreciated that the vocabulary used by the AI was tailored appropriately for students (Mean = 4.84) and noted that the app behaved consistently as expected

from an Android tool (Mean = 4.68). The visual appeal and professionalism of the interface were highly rated (Mean = 4.80). Most importantly, the experience of using the offline AI for active recall was found to be motivating (Mean = 4.68).

Discussion The overall UAT results validate the core hypothesis of this study: an offline, LAN-based AI system can effectively support cognitive learning. By processing data locally via OLLAMA, the CITL Assistant addresses the students' data privacy concerns while circumventing the "digital divide" associated with internet dependency. The transition from passive reading of dense digital textbooks to dynamic, context-aware questioning and interactive flashcards significantly enhances the students' active recall and knowledge retention.

RECOMMENDATIONS

To address the identified gaps the following recommendations are proposed:

Expand File Compatibility: future iterations should consider adding support for other formats such as word (.docx), plain text (.txt), and even image-based materials with the Optical Character Recognition (OCR). This would broaden usability across academic resources.

Integrate Cloud-based API(optional): the app can also use offline to online synchronization of API to broaden AI capabilities.

Offline Resources Sharing: Since the system utilize its offline security, adding LAN-based peer-to-peer sharing of study resources and quizzes could strengthen recall and retention.

Scalability and Maintenance: it is recommended to plan for future updates, bug fixes, and scalability to accommodate larger datasets and additional AI models.

Faculty Integration: it is recommended that for future iterations there should be a faculty dashboard wherein they will be able to manage all the learning materials for the student.

CONCLUSION

The development of the Cognitive Interactive Text Learner (CITL) Assistant successfully addressed the distinct cognitive learning challenges faced by IT students at CBSUA - Sipocot Campus. By leveraging Rapid Application Development (RAD) and integrating an offline LAN-based AI server (OLLAMA), the researchers delivered a secure, responsive, and interactive mobile study companion.

The application achieved its objectives by automating the creation of customized learning aids—such as summaries, interactive quizzes, and flashcards—directly from comprehensive PDF modules without relying on cloud APIs. The User Acceptance Test (UAT) results demonstrated excellent levels of Effectiveness (4.80), Efficiency (4.62), and Satisfaction (4.76) among target

users, confirming that the tool's interface is intuitive and its AI-generated content is highly relevant. Ultimately, the CITL Assistant proves that localized AI processing can effectively reduce cognitive load, transform passive reading habits into active learning workflows, and foster a more efficient, self-paced educational environment.

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